5G RAN

Layered QoS Feature Parameter Description

Issue Draft A

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the support for Layered QoS by the scenario where the core network transmits only the downlink PSIHI set to **False**. For details, see **4.1.1 PDU Set Identification**, **4.4.1.1 Data Preparation**, and **4.4.1.2 Using MML Commands**.	Added the **PSIHI_FALSE_ONLY_PROC_SW** option to the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Renamed the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter as **DL Layered QoS Switch**. For details, see **4.4.1.1 Data Preparation**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-091203	Layered QoS	**4 Layered QoS**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Layered QoS	Supported only in SA networking	**4 Layered QoS**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Layered QoS	Supported only in low frequency bands	**4 Layered QoS**

3 Overview

A QoS flow is the minimum granularity in the existing 5G QoS mechanism. In a QoS flow, all data packets are managed according to the same QoS rule, but the QoS management of a packet is done independently of other packets. However, in certain service scenarios, different packets in the same QoS flow may have varying QoS management requirements. In response to the varying requirements, 3GPP Release 18 introduces the concept of Protocol Data Unit (PDU) Set. According to 3GPP TS 23.501 V18.0.0, a PDU Set is a set of one or more application-layer PDUs that carry the same information unit (for example, a video frame or slice for XR). A PDU Set is also referred to as a data packet set.

To ensure low latency and high reliability for delay-sensitive services, Huawei introduces Layered QoS. This feature enables QoS handling based on PDU Set, which is a smaller granularity. With this feature, the core network identifies the characteristics of services (such as cloud VR and cloud gaming) and sends PDU Set information to the RAN. Based on the received information, the RAN precisely applies differentiated guarantee by taking many actions, including applying differentiated scheduling priorities, identifying frame importance, and discarding frames. This feature saves air interface resources and improves service experience.

4 Layered QoS

4.1 Principles

Table 4-1 lists the functions involved in Layered QoS. This feature is controlled by the **DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter. Layered QoS-related functions can take effect only after this option is selected.

Table 4-1 Functions involved in the Layered QoS feature

Function Name	Description	Chapter/Section
PDU Set identification	The gNodeB obtains the PDU Set and its identification information sent from the core network to identify the PDU Set mode, enhancing service identification accuracy. This is a basic function of other Layered QoS functions.	**4.1.1 PDU Set Identification**
Downlink PDU Set integrity protection	The gNodeB accurately detects the mapping of downlink frames with data packets and the frame size based on the PSSN, PSSize, and PSN fields sent from the core network. In this way, this function offers precise differentiated downlink scheduling, thereby improving service experience.	**4.1.2 Downlink PDU Set Integrity Protection**

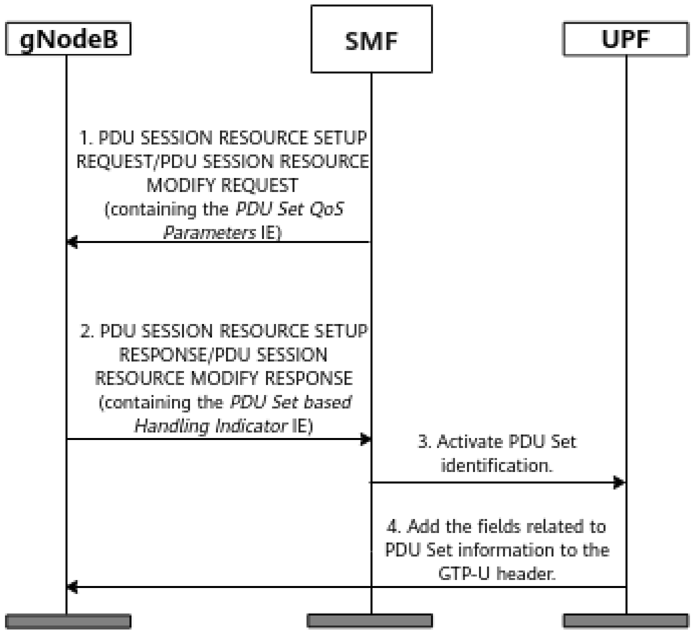
Function Name	Description	Chapter/Section
Downlink PSIHI-based discarding	The gNodeB identifies data packets in the same PDU Set based on the PSSN field sent from the core network. In the PDCP buffer of the gNodeB, if any data packet waits for a period longer than the discard timer, the gNodeB will discard the data packet and other data packets in the PDU Set to which the data packet belongs. This avoids wasting network resources.	**4.1.3 Downlink PSIHI-based Discarding**
Downlink PSI-based discarding	The gNodeB identifies the priority of each PDU Set based on the PSI field sent from the core network. In cases of cell congestion, the gNodeB preferentially discards low-priority PDU Sets in the downlink to preferentially guarantee the reception of high-priority PDU Sets at the receiver.	**4.1.4 Downlink PSI-based Discarding**
Other PDU Set-based downlink experience enhancements	To further improve downlink service experience, Layered QoS incorporates functions such as high-priority UE guarantee, intra-base-station CA downlink data split, and integrity of frame transmission during handovers. If a UE has a bearer configured with the R18 PDU Set mode, the corresponding function takes effect.	**4.1.5 Other PDU Set-based Downlink Experience Enhancements**

4.1.1 PDU Set Identification

PDU Set identification enables the gNodeB to obtain PDU Set information sent from the core network, enhancing service identification accuracy. **The user plane function (UPF) includes the PDU Set information in the GPRS tunneling protocol-user plane (GTP-U) header of a data packet sent to the gNodeB to indicate the PDU Set mode.** The size of a PDU Set is determined by the settings of the UPF in the core network. It can be less than or equal to a frame. This function applies only to the downlink. This function is controlled by the **gNBQciBearer**.***IdentificationMode*** parameter.

- If this parameter is set to **R18_PDU_SET_MODE**, this function is enabled.
- If this parameter is set to **PACKET_INTERVAL_MODE**, this function is disabled. In this case, the gNodeB determines the mode of each frame based on the packet interval.

Figure 4-1 Interaction between the gNodeB and core network



This function requires interaction between the gNodeB and core network, as shown in **Figure 4-1**. The procedure is as follows:

1. (Control plane) If the core network supports PDU Set identification, the session management function (SMF) includes the downlink PDU Set Delay Budget (PSDB) or downlink PDU Set Integrated Handling Information (PSIHI) in the *PDU Set QoS Parameters* IE of the "PDU SESSION RESOURCE SETUP REQUEST" or "PDU SESSION RESOURCE MODIFY REQUEST" message sent to the gNodeB.
2. (Control plane) If the gNodeB that supports PDU Set identification receives a message from the SMF, the gNodeB sends a "PDU SESSION RESOURCE SETUP RESPONSE" or "PDU SESSION RESOURCE MODIFY RESPONSE" message containing the *PDU Set based Handling Indicator* IE to the SMF. In this case, PDU Set identification takes effect. This function can be enabled for a gNodeB by setting the **gNBQciBearer**.***IdentificationMode*** parameter to **R18_PDU_SET_MODE**. If the SMF indicates that the *PDU Set QoS Parameters* IE contains only the downlink PSIHI (value: **False**) in the step described in **1**, the **PSIHI_FALSE_ONLY_PROC_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter must be selected for this function to take effect.
3. (Data plane) After receiving the message and determining that this function takes effect for the corresponding QoS flow, the SMF instructs the UPF to activate this function. The UPF adds the fields related to PDU Set information to the GTP-U header of each PDU. **Table 4-2** lists such fields.
4. (Data plane) The gNodeB parses the information in the GTP-U header, identifies each PDU Set, and performs subsequent differentiated handling.

Table 4-2 Fields related to PDU Set information

Field	Abbreviation	Description
PDU Set Sequence Number	PSSN	Indicates the SN of the PDU Set to which a PDU belongs. It is the identifier of the PDU Set.
PDU Sequence Number within a PDU Set	PSN	Indicates the SN of a PDU in a PDU Set.
End PDU of the PDU Set	EPDU	Indicates the last PDU in a PDU Set.
PDU Set Size	PSSize	Indicates the total size of all PDUs in the PDU Set to which a PDU belongs.
PDU Set Importance	PSI	Indicates the importance of a PDU Set relative to other PDU Sets in the same QoS flow. This field is indicated by an integer from 1 to 15. A smaller value indicates a higher importance.
PDU Set Size Indicator	PSSI	Indicates whether the PSSize field exists. The value **0** indicates that the PSSize field does not exist; the value **1** indicates that the PSSize field exists.

NOTE

In this document, a UE that has a bearer in R18 PDU Set mode is referred to as an R18 PDU Set UE.

4.1.2 Downlink PDU Set Integrity Protection

Using the PDU Set and related information transmitted from the core network, the gNodeB can perform more precise scheduling and transmission, enhancing service experience. To achieve this, downlink PDU Set integrity protection is introduced. **This function allows the gNodeB to accurately identify the mapping between the downlink PDU Set and data packets, PDU Set size, and other information. This is implemented through the PDU Set identification information such as PSSN, PSSize, and PSN obtained from the GTP-U header.** Based on the identified information, the gNodeB performs differentiated processing as follows:

- The gNodeB measures the delay for transmitting a PDU Set on a bearer in R18 PDU Set mode.

  If the delay exceeds the frame loss delay threshold, the PDU Set is discarded. The gNodeB periodically measures the proportion of discarded PDU Sets on high-priority bearers. A high proportion results in a downgrade from a high-priority bearer to a common bearer. The frame loss delay threshold is determined as follows: If the core network carries the downlink PSDB, it is the difference between the downlink PSDB and the value of the **gNBQciBearer**.***CoreNetDelayBudget*** parameter. If the core network does not carry the downlink PSDB, it equals the value of the **gNBQciBearer**.***CloudXFrmLossDelayThld*** parameter.

- The gNodeB measures the total downlink PRB usage of all high-priority bearers in R18 PDU Set mode in a cell.
  - If the total usage is less than the value of **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***, the gNodeB calculates the transmission delay for these bearers. If the transmission delay of a bearer does not exceed the maximum scheduling duration threshold, the gNodeB raises its downlink scheduling priority. The threshold is determined as follows: If the core network carries the downlink PSDB, it is the difference between the downlink PSDB and the value of the **gNBQciBearer**.***CoreNetDelayBudget*** parameter. If the core network does not carry the downlink PSDB, it equals the value of the **NRDUCellQciBearer**.***MaxCloudServSchTransTime*** parameter.
  - If the total usage is greater than or equal to the value of **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***, the gNodeB downgrades some high-priority bearers in R18 PDU Set mode to common bearers. This ensures that the total usage stays below the threshold.

Downlink PDU Set integrity protection is jointly controlled by the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter and the **CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter. This function is enabled when both of the following conditions are met:

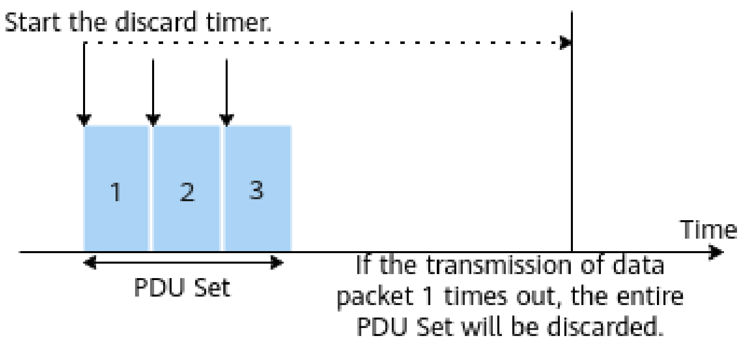
- For non-Cloud X bearers in R18 PDU Set mode, the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter is selected. Cloud X services refer to Cloud VR or Cloud Gaming services. For details, see "Cloud X UE identification" in *Ultimate Cloud X Experience*.
- The **CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter is selected.

4.1.3 Downlink PSIHI-based Discarding

If a packet in a downlink video frame is lost or damaged, the entire video frame typically fails to be displayed due to decoding errors. In such a case, transmitting the remaining data packets wastes network resources and affects service experience. Against this backdrop, downlink PSIHI-based discarding is introduced. PSIHI is one of the PDU Set QoS parameters sent from the core network to the gNodeB. This parameter indicates whether all data packets of a PDU Set are needed for the application layer of the receiver to parse the PDU Set. If this parameter is set to **True** for downlink, the gNodeB needs to transmit the entire PDU Set.

This function enables the gNodeB to identify data packets in the same PDU Set based on PSSN. After a data packet enters the PDCP buffer, the gNodeB will start a downlink PDCP discard timer. **In the PDCP buffer of the gNodeB, if any data packet waits for a period longer than the discard timer, the gNodeB will discard the data packet and other data packets in the PDU Set to which the data packet belongs, that is, the entire PDU Set. This avoids wasting network resources.** Figure 4-2 shows the principles of downlink PSIHI-based discarding.

Figure 4-2 Principles of downlink PSIHI-based discarding



This function takes effect only when both the following conditions are met:

- On the gNodeB, the function is enabled by selecting the **PSIHI_BASED_DISCARD_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter.
- On the core network, the downlink PSIHI is set to **True**.

The downlink PDCP discard timer is specified by the **gNBPdcpParamGroup**.***DlPdcpDiscardTimer*** parameter.

NOTE

- If the GTP-U header of a data packet does not contain the PSSN field, the gNodeB cannot determine whether the data packet belongs to the same PDU Set as other data packets. Therefore, this function does not take effect for such data packets.
- If the PSSN field in the GTP-U header of a data packet changes, the gNodeB considers that the entire PDU Set to which the data packet belongs has arrived. However, if data packets A and B in the same PDU Set are interrupted by data packet C in another PDU Set, the gNodeB considers that data packets A and B do not belong to the same PDU Set and does not discard the entire PDU Set.

4.1.4 Downlink PSI-based Discarding

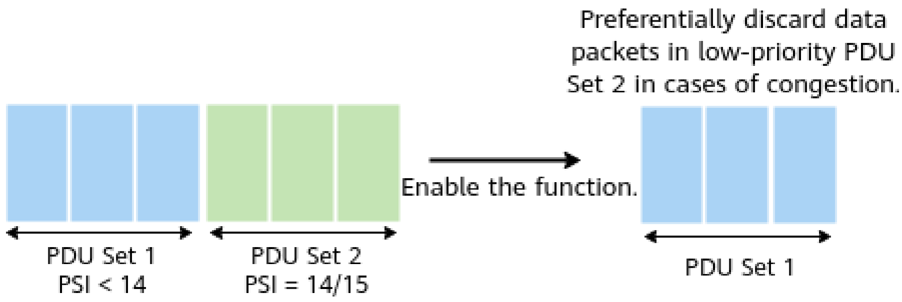
Downlink PSI-based discarding enables the gNodeB to identify transmission priorities of different data packets, allowing it to prioritize high-priority data transmission, thereby improving service experience. This function is controlled by the **PSI_BASED_DISCARD_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter. PSI is one of the PDU Set identifiers sent by the core network to the gNodeB through the GTP-U header. This identifier indicates the relative importance of a PDU Set in a QoS flow, with smaller values signifying higher importance.

This function shortens the discard timer for downlink low-priority PDU Sets. As shown in Figure 4-3, after this function is enabled, **the gNodeB identifies the**

priority of each PDU Set based on the PSI field sent from the core network. In cases of cell congestion, the gNodeB preferentially discards low-priority PDU Sets in the downlink to preferentially guarantee the reception of high-priority PDU Sets at the receiver. Specifically, PSI values **14** and **15** are defined as low priorities, and PSI values less than 14 are defined as high priorities. The principles of this function are as follows:

- Congested cells:
  - High-priority PDU Sets in the downlink: The downlink PDCP discard timer is used. In the PDCP buffer of the gNodeB, if any data packet in a high-priority PDU Set stays longer than the discard timer, the gNodeB will discard the data packet.
  - Low-priority PDU Sets in the downlink: The low-priority downlink discard timer, which is shorter, is used. In the PDCP buffer of the gNodeB, if any data packet in a low-priority PDU Set stays longer than the discard timer, the gNodeB will discard the data packet.
- Non-congested cells: In the PDCP buffer of the gNodeB, if any data packet in a PDU Set waits for a period longer than the downlink PDCP discard timer, the gNodeB will discard the data packet.

Figure 4-3 Principles of downlink PSI-based discarding



The low-priority downlink discard timer is specified by the **gNBPdcpParamGroup**.***DlDisTimerForLowImportance*** parameter, and the downlink PDCP discard timer is specified by the **gNBPdcpParamGroup**.***DlPdcpDiscardTimer*** parameter. The value of **gNBPdcpParamGroup**.***DlDisTimerForLowImportance*** must be less than that of **gNBPdcpParamGroup**.***DlPdcpDiscardTimer***. Enabling PDU Set identification is a prerequisite for enabling downlink PSI-based discarding. For details, see **4.1.1 PDU Set Identification**.

4.1.5 Other PDU Set-based Downlink Experience Enhancements

The cooperation between certain functions and Layered QoS is enhanced to further improve downlink service experience. Before enabling these functions, ensure that PDU Set identification has been enabled. For details, see **4.1.1 PDU Set Identification**.

Frame Delay-based Scheduling

After this function is enabled, the gNodeB checks whether a UE has a bearer in R18 PDU Set mode. If yes, the gNodeB computes the PDU Set delay for the bearer and predicts the remaining PDU Set transmission duration. A longer PDU Set delay results in a shorter remaining transmission duration, which in turn leads to a higher scheduling priority. This function is controlled by the **DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter. The **NRDUCellQciBearer**.***MaxCloudServSchTransTime*** parameter specifies the maximum transmission duration allowed by the frame-based scheduling algorithm.

High-Priority UE Guarantee

After this function is enabled, the gNodeB identifies all R18 PDU Set UEs. High-priority R18 PDU Set UEs are prioritized in resource allocation, followed by non-high-priority R18 PDU Set UEs. This function is controlled by the **CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter.

With this function in effect, the gNodeB preferentially guarantees resources for high-priority R18 PDU Set UEs. The scheduling priorities of these UEs can be adjusted by setting **gNBQciBearer**.***CloudXFrmLossDelayThld*** and **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***.

- For a single UE, the base station measures transmission delay of each frame of the R18 PDU Set UE.

  If the transmission delay of a frame is greater than the value of **gNBQciBearer**.***CloudXFrmLossDelayThld***, the frame is considered to be lost. The base station periodically measures the frame loss rate of high-priority R18 PDU Set UEs. If the frame loss rate of such a UE is excessively high, the base station degrades the UE to a common R18 PDU Set UE.
- The base station measures the total downlink PRB usage of high-priority R18 PDU Set UEs in a cell.
  - If this usage is less than the value of **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***, the frame delay-based scheduling algorithm is used only for high-priority R18 PDU Set UEs, but not for other R18 PDU Set UEs.
  - If this usage is greater than or equal to the value of **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***, some high-priority R18 PDU Set UEs will be treated as common R18 PDU Set UEs to ensure that this usage is less than the threshold.

Intra-Base-Station CA Downlink Data Split

Delay-sensitive services have stringent requirements on delay. However, in the existing CA data split architecture, data cannot be scheduled in a timely manner and as a result user experience requirements of such services cannot be met. To address this issue, intra-base-station CA downlink data split is introduced. This function is controlled by the **CLOUD_X_INTRA_GNB_CA_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter. After this function is enabled, the gNodeB checks whether a UE has a bearer in R18 PDU Set mode. If yes, this function takes effect for the UE. When the CA activation and data split thresholds are met, CA data split is started and frame delay-based scheduling is performed in the PCell and SCells. For details about the CA activation threshold, see section "SCell Configuration Procedure" in *Carrier Aggregation*. For details about the CA data split threshold, see the following conditions.

To ensure real-time performance, intra-base-station CA downlink data split can take effect only when all of the following conditions are met:

- The **CLOUD_X_INTRA_GNB_CA_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter is selected for both the PCell and SCells.
- The PRB usage of R18 PDU Set UEs measured in SCells is less than the PRB usage threshold for data split (specified by the **NRDUCellDlSch**.***CloudServSplitPrbThld*** parameter).
- The frame-based scheduling duration of R18 PDU Set UEs is less than the product of the data split duration proportion (specified by the **gNBQciBearer**.***CloudServSplitDurProp*** parameter) and the maximum transmission duration (specified by the **NRDUCellQciBearer**.***MaxCloudServSchTransTime*** parameter) allowed by the frame-based scheduling algorithm.
- The amount of data to be transmitted for R18 PDU Set UEs is greater than or equal to the product of the data split threshold (specified by the **gNBQciBearer**.***CloudServSplitCoef*** parameter) and the amount of data that can be scheduled per slot. A larger value of this parameter results in a lower probability for this function to take effect.

Integrity of Frame Transmission During Handovers

After this function is enabled, if a UE has a bearer in R18 PDU Set mode, the frame integrity handover algorithm is applied to the UE in handovers to improve the frame integrity in mobility scenarios. During a handover, source cell scheduling stops only after the frame transmission is complete. This function is controlled by the **CLOUD_X_FRAME_INTEGRITY_HO_SW** option of the **NRCellAlgoSwitch**.***CloudVrSwitch*** parameter.

User Resource Restriction

If excessive resources are allocated to R18 PDU Set UEs, other UEs will be compromised. To address this issue, UE resource restriction is introduced. This function restricts the downlink PRB resources of R18 PDU Set UEs.

After this function is enabled, the gNodeB adjusts the scheduling priority based on the downlink PRB usage of R18 PDU Set UEs. Preferential allocation of RBs to R18 PDU Set UEs is controlled by the **NRDUCellDlSch**.***CloudXResCtrlDlPrbThld*** parameter.

- If the downlink PRB usage of R18 PDU Set UEs is greater than or equal to this threshold, the base station does not apply the frame delay-based scheduling algorithm to such UEs but treat them as common UEs.
- If the downlink PRB usage of R18 PDU Set UEs is less than this threshold, the base station applies the frame delay-based scheduling algorithm to such UEs.

NOTE

- **NRDUCellQciBearer**.***MaxCloudServSchTransTime***: This parameter does not take effect when PDU Set identification is in effect and the core network carries the downlink PSDB parameter. In this case, the difference between the downlink PSDB and **gNBQciBearer**.***CoreNetDelayBudget*** will be used as the maximum allowed duration of transmission using the frame-based scheduling algorithm.
- **gNBQciBearer**.***CloudXFrmLossDelayThld***: This parameter does not take effect when PDU Set identification is in effect and the core network carries the downlink PSDB parameter. In this case, the difference between the downlink PSDB and **gNBQciBearer**.***CoreNetDelayBudget*** will be used as the frame loss delay threshold.

4.2 Network Analysis

4.2.1 Benefits

- PDU Set identification

  This function enhances the accuracy of service identification on the gNodeB. This function is a basic function for other Layered QoS functions and therefore does not provide any benefits.
- Downlink PDU Set integrity protection/Downlink PSIHI-based discarding/Downlink PSI-based discarding

  These functions produce the following benefits: (1) they decrease the 5QI-specific average downlink delay at the RLC layer; (2) they increase the proportion of 5QI-specific downlink buffer delay at the RLC layer below the delay threshold; and (3) they may increase the number of UEs with a specific 5QI in a cell if the 5QI-specific average downlink delay at the RLC layer meets service guarantee requirements.
  - 5QI-specific average downlink delay at the RLC layer = **N.RLC.DL.Cell5QI.Delay**/**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**
  - Proportion of 5QI-specific downlink buffer delay at the RLC layer below the delay threshold = **N.RLC.DL.TrfSDU.BfrDelay.DuCell5QI**/**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**. The aforementioned delay threshold is specified by the **gNBQciBearer**.***RlcDlDelayStatThld*** parameter and can be configured based on the delay requirements of different services.
- Other PDU Set-based downlink experience enhancements
  - Frame delay-based scheduling, high-priority UE guarantee, UE resource restriction, and integrity of frame transmission during handovers: After these functions are enabled, the user-perceived delay decreases and user experience improves.
  - Intra-base-station CA downlink data split: After this function is enabled, the user-perceived delay decreases and user experience improves. The benefits depend on the following factors:
    - Number of carriers: A larger number of carriers on the network and a larger bandwidth of each carrier result in greater benefits.
    - Network load: When the network is neither lightly loaded nor heavily loaded and the load of SCells remains unchanged, a heavier load of the PCell leads to greater benefits. When the load of the PCell

remains unchanged, a heavier load of SCells leads to smaller benefits.

- Coverage: A better coverage of SCells than that of the PCell brings greater benefits.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- PDU Set identification

  None
- Downlink PDU Set integrity protection

  None
- Downlink PSIHI-based discarding

  None
- Downlink PSI-based discarding

  None
- Other PDU Set-based downlink experience enhancements
  - Frame delay-based scheduling, high-priority UE guarantee, UE resource restriction, or integrity of frame transmission during handovers: When more R18 PDU Set UEs exist, they consume more downlink resources, causing less resources available for other UEs. As a result, the throughput of other services decreases.
  - Intra-base-station CA downlink data split
    - **Downlink Resource Block Utilizing Rate (DU)** in the PCell slightly decreases but **Downlink Resource Block Utilizing Rate (DU)** in the SCells slightly increases following data split.
    - When more R18 PDU Set UEs exist, they consume more downlink resources, causing less resources available for other UEs. As a result, the throughput of other services decreases.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-091203	Layered QoS	NR0S00LQS000	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- PDU Set identification

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Layered QoS	**DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter	*Layered QoS*	Layered QoS needs to be enabled for PDU Set identification to take effect.

- Downlink PDU Set integrity protection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	PDU Set identification needs to be enabled before the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Layered QoS	**DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter	*Layered QoS*	● Layered QoS is required for the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter to take effect. ● Cloud X service[a] or Layered QoS needs to be enabled before the **CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter is selected.
a: For details about Cloud X services, see *Ultimate Cloud X Experience*.				

- Downlink PSIHI-based discarding

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	PDU Set identification needs to be enabled before downlink PSIHI-based discarding is enabled.
Low-frequency TDD	Layered QoS	**DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter	*Layered QoS*	Layered QoS needs to be enabled for downlink PSIHI-based discarding to take effect.

- Downlink PSI-based discarding

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	PDU Set identification needs to be enabled before downlink PSI-based discarding is enabled.
Low-frequency TDD	Layered QoS	**DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter	*Layered QoS*	Layered QoS needs to be enabled for downlink PSI-based discarding to take effect.

- Other PDU Set-based downlink experience enhancements
  - Frame delay-based scheduling

    None
  - High-priority UE guarantee

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	High-priority UE guarantee takes effect only when the **gNBQciBearer**.***IdentificationMode*** parameter is set to **R18_PDU_SET_MODE** and the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter is selected.
Low-frequency TDD	Downlink PDU Set integrity protection	**INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter	*Layered QoS*	
Low-frequency TDD	Layered QoS	**DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch**.***IntelligentExpGuarSwitch*** parameter	*Layered QoS*	Cloud X service[a] or Layered QoS needs to be enabled before high-priority UE guarantee is enabled.

RAT	Function Name	Function Switch	Reference	Description
a: For details about Cloud X services, see *Ultimate Cloud X Experience*.				

– Intra-base-station CA downlink data split

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-gNodeB intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Intra-base-station CA downlink data split depends on downlink intra-gNodeB intra-band CA.
Low-frequency TDD	Downlink intra-gNodeB intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Intra-base-station CA downlink data split depends on downlink intra-gNodeB intra-FR inter-band CA.
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	Intra-base-station CA downlink data split takes effect only when the **gNBQciBearer**.***IdentificationMode*** parameter is set to **R18_PDU_SET_MODE** and the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter is selected.
Low-frequency TDD	Downlink PDU Set integrity protection	**INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter	*Layered QoS*	

– Integrity of frame transmission during handovers

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	PDU Set identification needs to be enabled for integrity of frame transmission during handovers to take effect.

– User resource restriction

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	User resource restriction takes effect only when the **gNBQciBearer**.***IdentificationMode*** parameter is set to **R18_PDU_SET_MODE** and the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter is selected.
Low-frequency TDD	Downlink PDU Set integrity protection	**INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter	*Layered QoS*	

4.3.2.2 Mutually Exclusive Functions

- PDU Set identification

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QoS Class Identifier	**gNBQciBearer**.***Qci***	*QoS Management*	When **gNBQciBearer**.***Qci*** is set to **1**, **2**, or **5**, PDU Set identification cannot be enabled.

- Downlink PDU Set integrity protection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X UE identification	**gNBQciBearer**.***ServiceType*** set to **EMBB_CLOUD_VR** or **EMBB_CLOUD_GAMING**	*Ultimate Cloud X Experience*	When Cloud X UE identification is enabled, the **INTEGRITY_GUARANTEE_ALGO_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter cannot be selected.

- Downlink PSI-based discarding
  None
- Downlink PSIHI-based discarding
  None
- Other PDU Set-based downlink experience enhancements
  None

4.3.2.3 Function Impacts

- PDU Set identification

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of IMS-DC-related QoS flow mapping to DRBs	**IMS_DC_QOS_MAP_DRB_PROT_SW** option of the **gNodeBAlgo**.***VoiceSw*** parameter	None	PDU Set identification does not take effect for bearers with protection of IMS-DC-related QoS flow mapping to DRBs in effect.

- Downlink PDU Set integrity protection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cloud X high-priority UE guarantee	**CLOUD_X_HIGH_PRI_UE_GTEE_SW** option of the **NRDUCellAlgoSwitch**.***CloudVrSchSwitch*** parameter	*Ultimate Cloud X Experience*	In a cell where both Cloud X and non-Cloud X bearers are configured, the downlink PDU Set integrity protection function can take effect for both bearers. This reduces the number of available downlink PRBs for Cloud X bearers, reducing the gains of Cloud X high-priority UE guarantee.

- Downlink PSIHI-based discarding

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink PSI-based discarding	**PSI_BASED_DISCARD_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter	*Layered QoS*	When both downlink PSI-based discarding and downlink PSIHI-based discarding are in effect in a congested cell, the following logic applies: ● In the PDCP buffer of the gNodeB, if any data packet in a high-priority PDU Set stays longer than the discard timer, the gNodeB will discard the PDU Set. ● In the PDCP buffer of the gNodeB, if any data packet in a low-priority PDU Set stays longer than the discard timer, the gNodeB will discard the PDU Set.

- Downlink PSI-based discarding

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink PSIHI-based discarding	**PSIHI_BASED_DISCARD_SW** option of the **gNBQciBearer**.***DlLayeredQosSwitch*** parameter	*Layered QoS*	When both downlink PSI-based discarding and downlink PSIHI-based discarding are in effect in a congested cell, the following logic applies: ● In the PDCP buffer of the gNodeB, if any data packet in a high-priority PDU Set stays longer than the discard timer, the gNodeB will discard the PDU Set. ● In the PDCP buffer of the gNodeB, if any data packet in a low-priority PDU Set stays longer than the discard timer, the gNodeB will discard the PDU Set.

- Other PDU Set-based downlink experience enhancements
  - Frame delay-based scheduling

    None
  - High-priority UE guarantee

    None
  - Intra-base-station CA downlink data split

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	**CA_PRFM_BOOST_BSD_ON_MD_OPT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*Multi-Frequency Smart Aggregation*	CA Performance Boost Based on Multi-dimensional Optimization aims to improve the UE scheduling efficiency and therefore increase the average downlink throughput of CA UEs. Intra-base-station CA downlink data split aims to ensure the delay and user experience that can be achieved after data split for R18 PDU Set UEs. Therefore, when CA Performance Boost Based on Multi-dimensional Optimization takes effect, the benefits provided by intra-base-station CA downlink data split decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed data volume–triggered SCell deactivation	**SPLIT_VOL_BASED_SCELL_DEACT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*Carrier Aggregation*	When both intra-base-station CA downlink data split and distributed data volume–triggered SCell deactivation are enabled, the following impacts arise: ● A larger value of the **gNBQciBearer**.***CloudServSplitCoef*** parameter results in a higher CA downlink data split threshold for R18 PDU Set UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of the **N.CA.SCell.Deact.Att** and **N.CA.SCell.Deact.Succ** counters will increase. ● A smaller value of the **gNBQciBearer**.***CloudServSplitCoef*** parameter results in a lower CA downlink data split threshold for R18 PDU Set UEs, a longer SCell-activated duration, and higher UE power consumption.

- Integrity of frame transmission during handovers

  None

- User resource restriction

  None

4.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

This function requires that the core network support functions related to PDU Set (introduced in 3GPP Release 18). Check that software versions for the core network are correct.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-3, **Table 4-4**, **Table 4-5**, **Table 4-6**, **Table 4-7**, **Table 4-8**, **Table 4-9**, **Table 4-10**, and **Table 4-11** describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation (PDU Set identification)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	Intelligent Experience Guarantee Switch	**NRCellFeatureSwitch**.***IntelligentExpGuarSwitch***	DL_LAYERED_QOS_SW	It is recommended that PDU Set identification be enabled for delay-sensitive services or for other services that require differentiated guarantee. PDU Set identification requires that this option be selected.
Low-frequency TDD	Identification Mode	**gNBQciBearer**.***IdentificationMode***	None	PDU Set identification requires that this parameter be set to **R18_PDU_SET_MODE**.
Low-frequency TDD	Service Type	**gNBQciBearer**.***ServiceType***	None	Set this parameter to the actual service type based on the network plan.
Low-frequency TDD	DL Layered QoS Switch	**gNBQciBearer**.***DlLayeredQosSwitch***	PSIHI_FALSE_ONLY_PROC_SW	PDU Set identification requires that this option be selected when the SMF indicates that the *PDU Set QoS Parameters* IE contains only the downlink PSIHI (value: **False**).

Table 4-4 Parameters used for activation (downlink PDU Set integrity protection)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	DL Layered QoS Switch	**gNBQciBearer**.***DlLayeredQosSwitch***	INTEGRITY_GUARANTEE_ALGO_SW	For non-CloudX bearers, downlink PDU Set integrity protection requires that this option be selected.
Low-frequency TDD	CloudX Sch Switch	**NRDUCellAlgoSwitch**.***CloudVrSchSwitch***	CLOUD_X_HIGH_PRI_UE_GTEE_SW	Downlink PDU Set integrity protection requires that this option be selected.
Low-frequency TDD	Core Network Delay Budget	**gNBQciBearer**.***CoreNetDelayBudget***	None	Downlink PDU Set integrity protection requires that this parameter be set if the core network carries the downlink PSDB. You are advised to set this parameter to a value recommended for a given QCI.
Low-frequency TDD	CloudX Frame Loss Delay Thld	**gNBQciBearer**.***CloudXFrmLossDelayThld***	None	Downlink PDU Set integrity protection requires that this parameter be set when the core network does not carry the downlink PSDB. You are advised to use the recommended value.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CloudX Resource Control DL PRB Threshold	**NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***	None	Downlink PDU Set integrity protection requires that this parameter be set. You are advised to set this parameter to **80**.
Low-frequency TDD	Max Cloud Service Scheduling Transfer Time	**NRDUCellQciBearer**.***MaxCloudServSchTransTime***	None	Downlink PDU Set integrity protection requires that this parameter be set when the core network does not carry the downlink PSDB. The value **10** is recommended.

Table 4-5 Parameters used for activation (downlink PSIHI-based discarding)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	DL Layered QoS Switch	**gNBQciBearer**.***DlLayeredQosSwitch***	PSIHI_BASED_DISCARD_SW	Downlink PSIHI-based discarding requires that this option be selected.
Low-frequency TDD	Downlink PDCP Discard Timer	**gNBPdcpParamGroup**.***DlPdcpDiscardTimer***	None	Downlink PSIHI-based discarding requires that this parameter be set. You are advised to set this parameter to a value recommended for a given QCI.

Table 4-6 Parameters used for activation (downlink PSI-based discarding)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	DL Layered QoS Switch	**gNBQciBearer**.***DlLayeredQosSwitch***	PSI_BASED_DISCARD_SW	Downlink PSI-based discarding requires that this option be selected.
Low-frequency TDD	DL Discard Timer for Low Importance	**gNBPdcpParamGroup**.***DlDisTimerForLowImportance***	None	Downlink PSI-based discarding requires that this parameter be set. You are advised to set this parameter to a value recommended for a given QCI.
Low-frequency TDD	Downlink PDCP Discard Timer	**gNBPdcpParamGroup**.***DlPdcpDiscardTimer***	None	Downlink PSI-based discarding requires that this parameter be set. You are advised to set this parameter to a value recommended for a given QCI.

Table 4-7 Parameters used for activation (frame delay-based scheduling)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	Intelligent Experience Guarantee Switch	**NRCellFeatureSwitch**.***IntelligentExpGuarSwitch***	DL_LAYERED_QOS_SW	Frame delay-based scheduling requires that this option be selected.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	Max Cloud Service Scheduling Transfer Time	**NRDUCellQciBearer**.***MaxCloudServSchTransTime***	None	Frame delay-based scheduling requires that this parameter be set when the core network does not carry the downlink PSDB. The value **10** is recommended.

Table 4-8 Parameters used for activation (high-priority UE guarantee)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	CloudX Sch Switch	**NRDUCellAlgoSwitch**.***CloudVrSchSwitch***	CLOUD_X_HIGH_PRI_UE_GTEE_SW	High-priority UE guarantee requires that this option be selected.
TDD	CloudX Frame Loss Delay Thld	**gNBQciBearer**.***CloudXFrmLossDelayThld***	None	High-priority UE guarantee requires that this parameter be set when the core network does not carry the downlink PSDB. You are advised to use the recommended value.
TDD	CloudX Resource Control DL PRB Threshold	**NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***	None	High-priority UE guarantee requires that this parameter be set. You are advised to set this parameter to **80**.

Table 4-9 Parameters used for activation (intra-base-station CA downlink data split)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CloudX Sch Switch	**NRDUCellAlgoSwitch**.***CloudVrSchSwitch***	CLOUD_X_INTRA_GNB_CA_ENH_SW	Intra-base-station CA downlink data split requires that this option be selected.
Low-frequency TDD	Cloud Service Split PRB Usage Threshold	**NRDUCellDlSch**.***CloudServSplitPrbThld***	None	Intra-base-station CA downlink data split requires that this parameter be set. You are advised to use the recommended value.
Low-frequency TDD	Cloud Service Split Duration Proportion	**gNBQciBearer**.***CloudServSplitDurProp***	None	Intra-base-station CA downlink data split requires that this parameter be set. You are advised to use the recommended value.
Low-frequency TDD	Cloud Service Split Coefficient	**gNBQciBearer**.***CloudServSplitCoef***	None	Intra-base-station CA downlink data split requires that this parameter be set. You are advised to use the recommended value.

Table 4-10 Parameters used for activation (integrity of frame transmission during handovers)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	CloudX Switch	**NRCellAlgoSwitch**.***CloudVrSwitch***	CLOUD_X_FRAME_INTEGRITY_HO_SW	Integrity of frame transmission during handovers requires that this parameter be selected.

Table 4-11 Parameters used for activation (user resource restriction)

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	CloudX Resource Control DL PRB Threshold	**NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***	None	User resource restriction requires that this parameter be set. You are advised to set this parameter to **80**.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- PDU Set identification

```
//Turning on the switch for downlink Layered QoS
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=DL_LAYERED_QOS_SW-1;
//Configuring the normal service (which is used as an example)
MOD GNBQCIBEARER: Qci=9, ServiceType=NORMAL;
//Setting IdentificationMode to R18_PDU_SET_MODE
MOD GNBQCIBEARER: Qci=9, IdentificationMode=R18_PDU_SET_MODE;
//(Optional) Selecting PSIHI_FALSE_ONLY_PROC_SW when the SMF indicates that the PDU Set QoS
```



```
Parameters IE contains only the downlink PSIHI (value: False)
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSIHI_FALSE_ONLY_PROC_SW-1;
```

- Downlink PDU Set integrity protection

```
//Turning on the switch for Cloud X high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-1;
//Turning on the switch for downlink PDU Set integrity protection for non-Cloud X bearers
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=INTEGRITY_GUARANTEE_ALGO_SW-1;
//Configuring the delay budget for the core network
MOD GNBQCIBEARER: Qci=9, CoreNetDelayBudget=2000;
//Configuring the threshold for the frame loss delay
MOD GNBQCIBEARER: Qci=9, CloudXFrmLossDelayThld=10;
//Configuring the maximum scheduling transmission duration
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, MaxCloudServSchTransTime=10;
//Configuring the downlink PRB usage threshold
MOD NRDUCELLDLSCH: NrDuCellId=0, CloudXResCtrlDlPrbThld=80;
```

- Downlink PSIHI-based discarding

```
//Turning on the switch for downlink PSIHI-based discarding
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSIHI_BASED_DISCARD_SW-1;
//Configuring the downlink PDCP discard timer
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=1, DlPdcpDiscardTimer=MS150;
```

- Downlink PSI-based discarding

```
//Turning on the switch for downlink PSI-based discarding
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSI_BASED_DISCARD_SW-1;
//Configuring the low-priority downlink discard timer
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=1, DlDisTimerForLowImportance=MS100;
//Configuring the downlink PDCP discard timer
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=1, DlPdcpDiscardTimer=MS150;
```

- Frame delay-based scheduling

```
//Turning on the switch for downlink Layered QoS
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=DL_LAYERED_QOS_SW-1;
//Configuring the maximum scheduling transmission duration
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, MaxCloudServSchTransTime=10;
```

- High-priority UE guarantee

```
//Turning on the switch for high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-1;
//Configuring the frame loss delay threshold
MOD GNBQCIBEARER: Qci=9, CloudXFrmLossDelayThld=10;
//Configuring the downlink PRB usage threshold
MOD NRDUCELLDLSCH: NrDuCellId=0, CloudXResCtrlDlPrbThld=80;
```

- Intra-base-station CA downlink data split

```
//Turning on the switch for intra-base-station CA downlink data split (assuming that cell 1 is the PCell
and cell 2 is an SCell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=2,
CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-1;
MOD NRDUCELLDLSCH: NrDuCellId=2, CloudServSplitPrbThld=50;
MOD GNBQCIBEARER: Qci=6, CloudServSplitDurProp=8, CloudServSplitCoef=2;
```

- Integrity of frame transmission during handovers

```
//Turning on the switch for integrity of frame transmission during handovers
MOD NRCELLALGOSWITCH: NrCellId=0, CloudVrSwitch=CLOUD_X_FRAME_INTEGRITY_HO_SW-1;
```

- User resource restriction

```
//Configuring the downlink PRB usage threshold for resource control
MOD NRDUCELLDLSCH: NrDuCellId=0, CloudXResCtrlDlPrbThld=80;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Downlink PDU Set integrity protection

```
//Turning off the switch for Cloud X high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-0;
//Turning off the switch for downlink PDU Set integrity protection for non-Cloud X bearers
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=INTEGRITY_GUARANTEE_ALGO_SW-0;
```

- Downlink PSIHI-based discarding

```
//Disabling PSIHI-based discarding
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSIHI_BASED_DISCARD_SW-0;
```

- Downlink PSI-based discarding

```
//Disabling PSI-based discarding
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSI_BASED_DISCARD_SW-0;
```

- Frame delay-based scheduling

```
//Turning off the switch for downlink Layered QoS
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=DL_LAYERED_QOS_SW-0;
```

- High-priority UE guarantee

```
//Turning off the switch for high-priority UE guarantee
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CloudVrSchSwitch=CLOUD_X_HIGH_PRI_UE_GTEE_SW-0;
```

- Intra-base-station CA downlink data split

```
//Turning off the switch for intra-base-station CA downlink data split
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=2,
CloudVrSchSwitch=CLOUD_X_INTRA_GNB_CA_ENH_SW-0;
```

- Integrity of frame transmission during handovers

```
//Turning off the switch for integrity of frame transmission during handovers
MOD NRCELLALGOSWITCH: NrCellId=0, CloudVrSwitch=CLOUD_X_FRAME_INTEGRITY_HO_SW-0;
```

- User resource restriction

```
//Configuring the downlink PRB usage threshold for resource control
MOD NRDUCELLDLSCH: NrDuCellId=0, CloudXResCtrlDlPrbThld=100;
```

- PDU Set identification

```
//(Optional) Deselecting PSIHI_FALSE_ONLY_PROC_SW when the SMF indicates that the PDU Set QoS
Parameters IE contains only the downlink PSIHI (value: False)
MOD GNBQCIBEARER: Qci=9, DlLayeredQosSwitch=PSIHI_FALSE_ONLY_PROC_SW-0;
//Setting IdentificationMode to PACKET_INTERVAL_MODE
MOD GNBQCIBEARER: Qci=9, IdentificationMode=PACKET_INTERVAL_MODE;
//Turning off the switch for downlink Layered QoS
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=DL_LAYERED_QOS_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- PDU Set identification

  After this function is enabled, run the **DSP NRCELLUENUMBER** command to query the UE statistics of an NR cell. If the value of **PDU Set Identification UE Number** is not **0**, this function has taken effect.

- Downlink PDU Set integrity protection/Downlink PSIHI-based discarding/Downlink PSI-based discarding

After this function is enabled, observe the decrease in the 5QI-specific average downlink delay at the RLC layer. For bearers where this function is enabled, a decrease in this indicator indicates that this function has taken effect. The indicator is calculated as follows: **N.RLC.DL.Cell5QI.Delay**/**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**.

- Frame delay-based scheduling/High-priority UE guarantee/Intra-base-station CA downlink data split/Integrity of frame transmission during handovers/User resource restriction

  After this function is enabled, you can observe the 5QI-specific average downlink delay at the MAC layer and 5QI-specific total downlink delay at the MAC layer to monitor the running status of this function. If any of them decreases, this function has taken effect. The two indicators are described as follows:

  – 5QI-specific average downlink delay at the MAC layer: **N.MAC.DL.Cell5QI.Delay**/**N.MAC.DL.Packets.Cell5QI**

  – 5QI-specific total downlink delay at the MAC layer: **N.MAC.DL.Cell5QI.Delay**

Table 4-12 Counters used in activation verification for Layered QoS

Counter ID	Counter Name
1911833755	**N.RLC.DL.Cell5QI.Delay**
1911822886	**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**
1911833754	**N.MAC.DL.Packets.Cell5QI**
1911833757	**N.MAC.DL.Cell5QI.Delay**

4.4.3 Network Monitoring

PDU Set identification is a basic function for other Layered QoS functions and therefore does not need to be monitored separately.

Downlink PDU Set integrity protection, downlink PSIHI-based discarding, and downlink PSI-based discarding can be monitored using the following indicators:

- Downlink delay: Monitor the 5QI-specific average downlink delay at the RLC layer and the proportion of 5QI-specific downlink buffer delay at the RLC layer below the delay threshold to evaluate the delay reduction.
- Cell capacity: If the 5QI-specific average downlink delay at the RLC layer meets service guarantee requirements, monitor the number of UEs configured with the 5QI in a cell to evaluate the capacity improvement.

The indicators listed in **4.2 Network Analysis** can be used to monitor the benefits and impacts of the following functions: frame delay-based scheduling, high-priority UE guarantee, intra-base-station CA downlink data split, integrity of frame transmission during handovers, and user resource restriction.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 23.501: "System architecture for the 5G System (5GS)"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Ultimate Cloud X Experience*
- *QoS Management*
- *License Control Item Lists*
- *Carrier Aggregation*
- *Multi-Frequency Smart Aggregation*

